3.1 *Theorem*:
$$\langle p|\,\hat{T} = \exp\left(\frac{-ibp}{\hbar}\right)\langle p|\,.$$

Proof. Given
$$\langle x|\,\hat{T} = \langle x-b|\,, \tag{1}$$
we can expand with
$$\begin{aligned}\langle p|\,\hat{T} &= \sum_x \langle p|x\rangle\,\langle x|\,\hat{T}\\ &= \int_{-\infty}^{\infty}\exp\left(\frac{-ipx}{\hbar}\right)\langle x-b|\,\mathrm{d}x\end{aligned}$$
and if we let $\xi = x-b$ and subsequently $x = \xi + b$ and $\mathrm{d}x = \mathrm{d}\xi$,
$$\begin{aligned}\langle p|\,\hat{T} &= \int_{-\infty}^{\infty}\exp\left(\frac{-ip(\xi+b)}{\hbar}\right)\langle \xi|\,\mathrm{d}\xi\\ &= \int_{-\infty}^{\infty}\exp\left(\frac{-ip\xi}{\hbar}\right)\exp\left(\frac{-ipb}{\hbar}\right)\langle \xi|\,\mathrm{d}\xi\\ &= \exp\left(\frac{-ipb}{\hbar}\right)\int_{-\infty}^{\infty}\exp\left(\frac{-ip\xi}{\hbar}\right)\langle \xi|\,\mathrm{d}\xi\end{aligned}$$
and the second term is an equivalent definition of $\langle p|$ as ξ lies in position-space, simplifying to
$$\langle p|\,\hat{T} = \exp\left(\frac{-ipb}{\hbar}\right)\langle p|\,.$$
□

3.2 *Theorem*:
$$\langle p|\,\hat{D} = \frac{ip}{\hbar}\langle p|\,.$$

Proof. Given the definition
$$\langle x|\,\hat{D} = \frac{\mathrm{d}}{\mathrm{d}x}\langle x|\,, \tag{2}$$
we can once again expand
$$\begin{aligned}\langle p|\,\hat{D} &= \sum_x \langle p|x\rangle\,\langle x|\,\hat{D}\\ &= \int_{-\infty}^{\infty}\exp\left(\frac{-ipx}{\hbar}\right)\frac{\mathrm{d}}{\mathrm{d}x}\langle x|\,\mathrm{d}x\end{aligned}$$

Using integration by parts, we can set

$$u = \exp\left(-\frac{ipx}{\hbar}\right) \qquad \mathrm{d}u = -\frac{ip}{\hbar}\exp\left(\frac{-ipx}{\hbar}\right)\mathrm{d}x$$
$$\mathrm{d}v = \frac{\mathrm{d}}{\mathrm{d}x}\langle x|\,\mathrm{d}x \qquad v = \langle x|$$

yielding the equation

$$\begin{aligned}\int_{-\infty}^{\infty}\exp\left(\frac{-ipx}{\hbar}\right)\frac{\mathrm{d}}{\mathrm{d}x}\langle x|\,\mathrm{d}x &= \exp\left(-\frac{ipx}{\hbar}\right)\langle x|\Big|_{-\infty}^{\infty} - \int_{-\infty}^{\infty}-\frac{ip}{\hbar}\exp\left(\frac{-ipx}{\hbar}\right)\langle x|\,\mathrm{d}x \\ &= \exp\left(-\frac{ipx}{\hbar}\right)\langle x|\Big|_{-\infty}^{\infty} + \frac{ip}{\hbar}\int_{-\infty}^{\infty}\exp\left(\frac{-ipx}{\hbar}\right)\langle x|\,\mathrm{d}x\end{aligned}$$

As $\lim_{x\to\pm\infty}\exp(-ipx/\hbar) = 0$, both terms simplify to

$$\langle p|\,\hat{D} = \frac{ip}{\hbar}\langle p|\,.$$

□

3.3 The mirror reflection operator $\hat{R}$ produces the reflection about the origin of any function in x-space. It is called the parity operator in physics. Its x-space instruction is $\langle x|\,\hat{R} = \langle -x|$. What is its p-space instruction?

Suppose we have a state ψ,

$$\begin{aligned}\langle p|\,\hat{R}\,|\psi\rangle &= \sum_x \langle p|x\rangle\,\langle x|\,\hat{R}\,|\psi\rangle \\ &= \int_{-\infty}^{\infty}\exp\left(-\frac{ipx}{\hbar}\right)\langle -x|\psi\rangle\,\mathrm{d}x\end{aligned}$$

Now let $\xi = -x$, producing

$$\begin{aligned}\langle p|\,\hat{R}\,|\psi\rangle &= -\int_{-\infty}^{\infty}\exp\left(\frac{ip\xi}{\hbar}\right)\langle \xi|\psi\rangle\,\mathrm{d}\xi \\ &= -\int_{\infty}^{-\infty}\exp\left(-\frac{i(-p)\xi}{\hbar}\right)\langle \xi|\psi\rangle\,\mathrm{d}\xi \\ &= \langle -p|\psi\rangle\end{aligned}$$

3.4-1 Given $\hat{\tau}\,|p\rangle = |p+Q\rangle$ where $Q = h/L$. *Theorem*:

$$\langle p|\,\hat{\tau} = \langle p-Q|\,.$$

Proof. $\langle p|\,\hat{\tau} = (\hat{\tau}\,|p\rangle)^\star = \langle p|\,\hat{\tau}^\dagger = \langle p-Q|$ □

3.4-2 *Theorem*:

$$\langle x|\,\hat{\tau} = \exp\left(\frac{iQx}{\hbar}\right)\langle x|$$

Proof.

$$\langle x| \hat{\tau} = \sum_p \langle x|p\rangle \langle p| \hat{\tau} = \sum_p (\langle p|x\rangle)^\star \langle p| \hat{\tau}$$
$$= \sum_p \exp\left(\frac{ipx}{\hbar}\right) \langle p - Q|$$

Now let $p' = p - Q$,

$$\langle x| \hat{\tau} = \sum_p \exp\left(\frac{i(p' + Q)x}{\hbar}\right) \langle p'|$$
$$= \sum_p \exp\left(\frac{ip'x}{\hbar}\right) \exp\left(\frac{iQx}{\hbar}\right) \langle p'|$$
$$= \exp\left(\frac{iQx}{\hbar}\right) \sum_p \exp\left(\frac{ip'x}{\hbar}\right) \langle p'|$$
$$= \exp\left(\frac{iQx}{\hbar}\right) \langle x|$$

□

3.5-1 *Theorem*:

$$\langle x| \hat{\tau}\hat{D} |\psi\rangle = \exp\left(\frac{iQx}{\hbar}\right) \frac{\mathrm{d}\psi}{\mathrm{d}x}$$

Proof. $\langle x| \hat{\tau}\hat{D} |\psi\rangle = \frac{\mathrm{d}}{\mathrm{d}x} \langle x| \hat{\tau} |\psi\rangle = \exp\left(\frac{iQx}{\hbar}\right) \frac{\mathrm{d}}{\mathrm{d}x} \langle x|\psi\rangle$. □

7.1 *Theorem*: $\hat{T} = e^{-b\hat{D}}$

Proof.

$$\langle p| \exp\left(-b\hat{D}\right) = \langle p| \exp(-bD(p))$$

where $D(p) = ip/\hbar$ is the eigenvalue of $\hat{D}$ for $\langle p|$, and knowing that

$$\langle p| \hat{T} = \exp\left(\frac{-ibp}{\hbar}\right) \langle p| \tag{3}$$

from problem 3.1,

$$\langle p| \hat{T} = \exp(-bD(p)) \langle p| = \exp\left(-b\hat{D}\right) \langle p| ,$$

thus

$$\hat{T} = e^{-b\hat{D}}.$$

□

7.2 *Theorem*: $\langle x|\,\hat{p}\,|\psi\rangle = \langle x|\,\hat{p}\hat{T}\,|\psi\rangle$

Proof. Given that $\hat{T} = e^{-b\hat{D}}$ per problem 7.1, $\langle x|\psi\rangle$ and $\langle x|\,\hat{T}\,|\psi\rangle$ differ by only the phase $e^{-b\hat{D}}$ which cannot have any effect on the probability as $\forall\theta, |e^{i\theta}|^2 = 1$. □

7.3 *Theorem*: $\forall\psi, \forall t,\ |\,\langle E|\psi\rangle\,|^2 = |\,\langle E|\,\hat{U}_t\,|\psi\rangle\,|^2$.

Proof. $\hat{U}_t = \exp\left(\frac{-it\hat{H}}{\hbar}\right)$ is a unit vector on the complex plane, so it cannot have any effect on the probability. □

7.4 What is the characteristic frequency $\Omega/2\pi$ for a neutron in a magnetic field of $1\,\mathrm{kG}$? In what region of the electromagnetic spectrum is this: radio, microwave, infrared, visible, or x-ray?

The characteristic frequency is

$$\omega = \frac{B}{2\pi}|\Omega| = \frac{B}{2\pi}\left|\frac{g_n e}{2m_n}\right| = \frac{1\times 10^3\,\mathrm{kG}}{2\pi}\left|\frac{(-3.83)(1.60\times 10^{-19}\,\mathrm{C})}{2(1.67\times 10^{-27}\,\mathrm{kg})}\right| \simeq 2.92\,\mathrm{MHz}$$

and this is a radiowave.

7.5 *Theorem*: $\hat{U}_t\,|\mu = 1/2\rangle = \frac{\hat{U}_t}{\sqrt{2}}(|m = 1/2\rangle + i\,|m = -1/2\rangle)$ where μ and m are eigenvalues of $\hat{S}_y$ and $\hat{S}_z$ respectively.

Proof. The unit time transform can be removed from both sides, so the proof is just

$$|\mu = 1/2\rangle = \frac{1}{\sqrt{2}}\,|m = 1/2\rangle + \frac{i}{\sqrt{2}}\,|m = -1/2\rangle$$

and this can be proven using the change-of-basis matrix from $\mathrm{eig}_\lambda\hat{S}_y$ to $\mathrm{eig}_\lambda\hat{S}_z$. This can be produed using the half-integer spinor rotation operator

$$\hat{R}(\theta, \hat{n}) = \exp\left(-\frac{i\theta}{2}\vec{\sigma}\cdot\hat{n}\right) \tag{4}$$

where θ is the rotation angle, $\hat{n}$ is the axis of rotation (in this case, the x-axis), and $\vec{\sigma} = (\sigma_x, \sigma_y, \sigma_z)$ is a vector of the Pauli matrices for each axis. For a half-integer spin on the x-axis, this simplifies (via a Taylor expansion not shown) to

$$\hat{R}_x(\theta) = I\cos\frac{\theta}{2} - i\sigma_x\sin\frac{\theta}{2} = \begin{bmatrix}\cos\frac{\theta}{2} & -i\sin\frac{\theta}{2}\\ -i\sin\frac{\theta}{2} & \cos\frac{\theta}{2}\end{bmatrix}. \tag{5}$$

Thus, given that we want a counterclockwise rotation from the y-axis to the z-axis, we need to compute $\hat{R}_x(-\pi/2)$, which is

$$\hat{R}_x(-\pi/2) = \frac{1}{\sqrt{2}}\begin{bmatrix}1 & i\\ i & 1\end{bmatrix}$$

and then, taking $|\mu = 1/2\rangle$ to be $(1\ 0)^T$ in the y-basis, we can use $\hat{R}_x(-\pi/2)$ as the change-of-basis matrix from the y-basis to the z-basis as

$$\hat{R}_x(-\pi/2)\begin{pmatrix}1\\0\end{pmatrix} = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\i\end{pmatrix},$$

which, written out, is equivalent to

$$|\mu = 1/2\rangle = \frac{1}{\sqrt{2}}\,|m = 1/2\rangle + \frac{i}{\sqrt{2}}\,|m = -1/2\rangle\,.$$

□

7.6 A neutron spinning initially in the $+y$-direction, $(\mu = 1/2)$, spends time t in a z-direction magnetic field of strength B. What is the probability of finding the initial spin $(\mu = 1/2)$ at time t?

This probability boils down to computing $||\,\langle\mu = 1/2|\,\hat{U}_t\,|\mu = 1/2\rangle\,||^2$, so all we need to compute is $\langle\mu = 1/2|\,\hat{U}_t\,|\mu = 1/2\rangle$. First, the time-evolution operator can be expanded to

$$\hat{U}_t = \exp\left(\frac{-it\hat{H}}{\hbar}\right) = \exp\left(\frac{-it\Omega\hat{S}_z}{\hbar}\right)$$

And given that the hamiltonian can only operate on states defined in the z-axis, the time-evolved operator can be expressed using equation 7.13 as

$$U_t\left|\mu = \frac{1}{2}\right\rangle = \frac{1}{\sqrt{2}}\exp\left(\frac{-it\Omega}{2}\right)\left|m = \frac{1}{2}\right\rangle + \frac{i}{\sqrt{2}}\exp\left(\frac{it\Omega}{2}\right)\left|m = -\frac{1}{2}\right\rangle$$

and similarly to the procedure described in the previous problem, we can then calculate $\langle\mu = 1/2|\,U_t\,|\mu = 1/2\rangle$ by performing a rotation by $\theta = 1/2$ along the x-axis, i.e., we can use $\hat{R}_x(\pi/2)$ as the $z \to y$ change-of-basis matrix:

$$\hat{R}_x(\pi/2) = \frac{1}{\sqrt{2}}\begin{bmatrix}1 & -i\\ -i & 1\end{bmatrix}$$

Thus, expressing $U_t\,|\mu = 1/2\rangle$ in the z-basis yields

$$\hat{U}_t\,|\mu = 1/2\rangle \equiv \frac{1}{\sqrt{2}}\begin{pmatrix}e^{-it\Omega/2}\\ ie^{it\Omega/2}\end{pmatrix}$$

and if we then perform the rotation into the y-axis using $\hat{R}_x(\theta/2)$

$$\hat{R}_x(\pi/2)\hat{U}_t\,|\mu = 1/2\rangle = \frac{1}{2}\begin{pmatrix}\lambda + \lambda^{-1}\\ i(\lambda - \lambda^{-1})\end{pmatrix}$$

where $\lambda = e^{it\Omega/2}$, or equivalently,

$$\hat{R}_x(\pi/2)\hat{U}_t\,|\mu = 1/2\rangle = \begin{pmatrix}\cosh(it\Omega/2)\\ i\sinh(it\Omega/2)\end{pmatrix} = \begin{pmatrix}\cos(t\Omega/2)\\ i\sin(t\Omega/2)\end{pmatrix},$$

and thus, the raw amplitude is

$$\langle \mu = 1/2 | \, U_t \, | \mu = 1/2 \rangle = \cos(t\Omega/2)$$

and the probability is

$$|\langle \mu = 1/2 | \, U_t \, | \mu = 1/2 \rangle|^2 = \cos^2(t\Omega/2) = \frac{1}{2}(1 + \cos \Omega t)$$

by the double-angle formula.

7.7 *Theorem*: $\exp\left(\frac{-it\hat{H}}{\hbar}\right) = 1 - \frac{it\hat{H}}{\hbar}$ for a stationary state where $E < \hbar/t$.

Proof. This can be proven quickly with a simple Maclaurin expansion. The Maclaurin expansion of e^x is

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2} + \dots,$$

$\square$

so therefore,

$$\exp\left(\frac{-it\hat{H}}{\hbar}\right) = 1 - \frac{it\hat{H}}{\hbar} + \frac{1}{2}\left(\frac{-it\hat{H}}{\hbar}\right)^2 + \dots$$

and as $E < \hbar/t$, then $t\hat{H}/h << 1$, so the only two important terms are the first two, meaning that

$$\exp\left(\frac{-it\hat{H}}{\hbar}\right) \simeq 1 - \frac{it\hat{H}}{\hbar}$$

7.8 *Theorem*: $\langle A \rangle = \langle \psi | \, \hat{A} \, | \psi \rangle \; \forall A, \forall \psi$.

Proof.

$$\begin{aligned}
\langle \psi | \, \hat{A} \, | \psi \rangle &= \sum_A \langle \psi | A \rangle \langle A | \, \hat{A} \, | \psi \rangle & &= \sum_A A \langle \psi | A \rangle \langle A | \psi \rangle \\
&= \sum_A A \langle A | \psi \rangle^\star \langle A | \psi \rangle & &= \sum_A A |\langle A | \psi \rangle|^2 \\
&= \sum_A A P(A|\psi) & &= \langle A \rangle
\end{aligned}$$

$\square$